RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Friday, 25 September 2026, 2:28 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:29 PM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void findIntersection(int arr1[], int n1, int arr2[], int n2) {
    int first = 1;
    // O(m*n) nested loop approach to find common elements
    for (int i = 0; i < n1; i++) {
        for (int j = 0; j < n2; j++) {
            if (arr1[i] == arr2[j]) {
                if (!first) {
                    printf(" ");
                }
                printf("%d", arr1[i]);
                first = 0;
                break; // Move to the next element in arr1 to avoid duplicate prints
            }
        }
    }
    printf("\n");
}

int main() {
    int t;
```

```
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        int n1, n2;

        // Read first array size and its elements
        if (scanf("%d", &n1) != 1) break;
        int arr1[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &arr1[i]);
        }

        // Read second array size and its elements
        if (scanf("%d", &n2) != 1) break;
        int arr2[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &arr2[i]);
        }

        // Find and print intersection
        findIntersection(arr1, n1, arr2, n2);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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